

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Which of the following was not cited as evidence for life in the martian meteorite ALH84001?' is 'Magnetite grains similar to those formed by bacteria on Earth'.
2. True or False: The correct answer to 'According to Kant, the basis of morality is the concept of:' is 'freedom.'.
3. True or False: The correct answer to 'Aquinas argues that the end of every action is:' is 'some emotional state.'.
4. True or False: The correct answer to 'As Ethanol burns it expels what into the ozone' is 'CO₂'.
5. True or False: The correct answer to 'Three half-lives after an isotope is prepared,' is '12.5% of the isotope decayed'.
6. True or False: The correct answer to 'Which American colony known for its religious tolerance did Roger Williams found in 1636?' is 'Vermont'.
7. True or False: The correct answer to 'when fuel efficiency increases, gasoline use in cars will' is 'decrease'.
8. True or False: The correct answer to 'What do predators live near in the same environment?' is 'Their dens'.
9. True or False: The correct answer to 'Of the following compounds, which is LEAST likely to behave as a Lewis acid?' is 'SCl₂'.
10. True or False: The correct answer to 'Opposite the winter solstice, the summer solstice features' is 'brighter fires'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Write a python function to find the next perfect square greater than a given number.
2. Write a python function to find the first repeated character in a given string.

End of Paper